

Autonomous Software Engineer & Maintainer

Codebase Mapping, Bug Investigation, Test Generation & Safe PR Creation

Scenario Brief: Continuous Code Maintenance for SaaSScale Inc.

Objective: Triage Sentry error reports, navigate complex codebases, write reproducing unit tests, fix the underlying bug, and submit ready-to-review Pull Requests.

● ● ● Autonomous

23/23 TESTS PASSED (74s)

REPO MAP (34ms):
src/analytics.py
src/revenue_calc.py
tests/test_revenue.py
Branch: fix/mom-zero-div

FAIL: test_mom_growth_zero_prior_month (ZeroDivisionError)

> Patched: revenue_calc.py line 84 (defensive 0-check)

> pytest tests/ -v: 23 passed in 1.42s

> GitHub PR #142 opened: "fix: guard MoM revenue calculation against 0 sales"

THE 5-STEP CODE ENGINEERING LOOP

- Repo Mapping:** Parses AST symbols across 240+ files in 34ms.
- Bug Reproduction:** Generates standalone test in isolated sandbox.
- Targeted Code Modification:** Edits source files via `file_edit`.
- Automated Test Execution:** Verifies 100% tests pass cleanly.
- Git Branch & PR Creation:** Opens structured GitHub PR for review.

REAL EMPIRICAL CASE (FROM FATHOM TEST SUITE)

In live case study 5 (05-case-studies.md), an autonomous Fathom engineer was tasked with building a complete Python CLI with MoM revenue analytics, sample data, and pytest coverage, achieving **23/23 passing tests** completely autonomously in 74 seconds.

Developer Superpower: Senior engineers focus on high-level system architecture while digital coworkers handle routine bugs, dependency upgrades, and test coverage.

Fathom Autonomous Workforce Runtime · Elastic-2.0 License · Strategic Whitepaper Presentation

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